F78AB - Actuarial and Financial Mathematics B
Computer Lab 2 – Stochastic interest rate models

Lemma 0.1 (Inverse Transform)**.** *Let X be a random variable with continuous distribution function Φ. Then the random variable $\Phi(X) \sim UNIF[0,1]$.*

Proof. As Φ is a distribution function, it takes values only in $[0,1]$. Fix $u \in [0,1]$. Consider the distribution function of $\Phi(X)$.

$$\mathbb{P}\left[\Phi(X) \leq u\right] = \mathbb{P}\left[X \leq \Phi^{-1}(u)\right] = \Phi\left(\Phi^{-1}(u)\right) = u,$$

i.e. $\Phi(X)$ has the distribution function of UNIF$[0,1]$. □

To apply the Inverse Transform Sampling Method.

To find a sample from a random variable X with continuous distribution function Φ:

1. Generate a sample u from a random variable $U \sim UNIF[0,1]$.
2. Calculate $x = \Phi^{-1}(u)$.
3. Then x is a sample of X.

The idea is that there is a future state of the world ω in which $u = U(\omega)$. Then, by the Lemma, $U(\omega) = \Phi(X(\omega))$. We are finding $X(\omega)$ by calculating $X(\omega) = \Phi^{-1}(U(\omega))$.

Example: Sampling from a standard normal random variable $Z \sim N(0,1)$ in Excel.

To sample from a standard normal random variable $Z \sim N(0,1)$	**Corresponding Excel commands**
1. Generate a sample u from a random variable $U \sim UNIF[0,1]$.	$= RAND()$
2. Calculate $z = \Phi^{-1}(u)$.	$= NORM.S.INV(\textit{Excel cell reference})$
3. Then z is a sample of Z.	

Question 1

The investment return earned in year n is denoted i_n. Assume $(1 + i_n)$ is lognormally distributed with parameters $\mu = 0.0867$ and $\sigma = 0.15$ for each $n = 1, 2, \ldots$.

You are to create a spreadsheet to generate samples of the investment returns over 10 years, using the Inverse Transform Sampling Method.

Above each column, you should show each of the following summary statistics: COUNTA, MEDIAN, AVERAGE, STDEV.S, MIN, MAX.

Do the following in a spreadsheet.

1. Generate 250 random numbers from $U(0, 1)$, using RAND(), in the same column. These will be used for Year 1.

2. Generate nine more columns of random numbers, to use in Years 2-10.

3. For Year 1, generate 250 samples from $Z \sim N(0, 1)$ using the Inverse Transform Sampling Method.

4. Generate nine more columns of random normal numbers.

5. Using the generated 250 normal samples, generate 250 samples from the random variable $1 + i_1$.

   Hint: Use the fact that $\ln(1 + i_1) = \mu + \sigma Z$, for some random variable $Z \sim N(0, 1)$.

6. Generate nine more columns of $1 + i_n$.

7. For each sample, calculate the amount that £10 000 will accumulate to, using the ten random returns that you have generated.

8. Based on the samples, calculate the (empirical) probability that £10 000 accumulates to more than £25 000 at the end of 10 years.

9. Based on the samples, calculate the (empirical) probability that £10 000 accumulates to less than £5 000 at the end of 10 years.

10. Based on the samples, calculate the (empirical) median accumulation at the end of 10 years.

11. Based on the samples, calculate the (empirical) mean accumulation at the end of 10 years.

Question 2

The effective rate of interest earned in the year from time $t - 1$ to t is denoted by i_t. It is assumed that $i_1, i_2, \ldots$ are independent and identically distributed, with $\mathbb{E}[i_t] = 0.08$, $\mathbb{E}[i_t^2] = 0.042$ and $\mathbb{E}[i_t^3] = 0.013$.

Let A_n represent the accumulation up to time n of £1 payable at times $0, 1, \ldots, n - 1$.

Using a spreadsheet, calculate $E[A_{10}]$, $E[A_{10}^2]$, $\text{var}(A_{10})$ and $E[A_{10}^3]$.

Note: Based on Q5 of Tutorial 2.